

Dev Patel

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Portfolio: <https://devpatel5804.github.io/portfolio/>

Back-End .NET Developer & Full-Stack Engineer specializing in **cloud-native systems**, **industrial automation**, and **scalable APIs**. Proven track record of architecting real-time monitoring platforms and data pipelines for **500+** renewable energy sites.

Education

Bhagwan Mahavir University, Surat, Gujarat, India

2021 – 2025

Bachelor of Technology in Computer Science

CGPA: 7.65 / 10.0 (SGPA: 9.00)

Experience

Adaptive Engineering Pvt. Ltd., Ahmedabad, Gujarat, India

Back-End .NET Developer

2025 – Present

- Architected a scalable cloud-integration layer bridging the gap between local industrial automation and unified cloud monitoring to aggregate distributed SCADA telemetry into a single-pane-of-glass CMS.
- Engineered robust data transition paths to AWS, transforming complex, multi-site telemetry into a seamless, real-time experience across CMS, AMS, and EMS platforms.
- Developed and optimized RESTful APIs to fetch, process, and stream this synchronized cloud data, delivering high-speed, real-time responses to the front-end user interface.
- Utilized Redis caching and Python scripts to optimize backend data processing pipelines, reducing system latency and database load during peak generation hours.

Neptune Infotech, Surat, Gujarat, India

Full-Stack Web Developer

2024 – 2025

- Engineered and deployed production-ready Laravel APIs integrated seamlessly with a React administrative panel for live production systems.
- Built core security and commerce features, including custom OTP verification, Google OAuth secure login, and Razorpay payment gateway integration.
- Designed and managed MySQL database structures to support high-performance, real-time data handling for connected mobile applications.

Projects

CMS (Central Monitoring System) – Multi-Site Renewable Energy Platform

Tech: .NET Core, AWS, Redis, PostgreSQL, React, SCADA

- Engineered a comprehensive, cloud-native Central Monitoring System (CMS) designed for a large-scale renewable energy portfolio, consolidating real-time telemetry from hundreds of distributed sites into a unified, actionable dashboard.
- Developed a Geo-Location engine to track and monitor 500+ sites across India on a single interactive map, providing instant health status and connectivity metrics.
- Architected a predictive engine using custom forecasting models to project Active Power, Radiation (GII), and Energy Generation for proactive grid management and data-driven scheduling.
- Built advanced data visualization tools, including Heatmap Charts for string monitoring, Generation vs. Target comparisons, and real-time Inverter Efficiency tracking across thousands of hardware units.
- Integrated Battery Energy Storage System (BESS) analysis, featuring state-of-charge (SoC) tracking and revenue-based Arbitrage Strategy modeling to optimize battery dispatch.
- Engineered Scorecard and Shortfall performance modules to monitor business-critical KPIs including CUF (Capacity Utilization Factor), PR (Performance Ratio), and Revenue Forecasting.
- Designed the scalable AWS data backbone to handle secure transition from local SCADA protocols to a centralized cloud environment, ensuring zero data loss across CMS, AMS, and EMS modules.

Advanced AI & ML Predictive Analytics Module | Edunet Foundation

Tech: Python, Machine Learning, Predictive Modeling

2023 – 2024

- Developed predictive models and data pipelines as part of an advanced AI/ML specialization.
- Analyzed complex datasets to extract actionable insights, sharpening skills in critical thinking, algorithm selection, and data preprocessing.

Technical Skills

- **Languages:** C#, JavaScript, Python, SQL, HTML, CSS
- **Frameworks & Libraries:** .NET Core, ASP .NET, Entity Framework, Laravel, React, Express, Bootstrap
- **Databases, Caching & Tools:** PostgreSQL, Redis, SQLite, MySQL, SSMS
- **Cloud & DevOps:** AWS (S3, EC2, Lambda, DynamoDB), Azure, Docker, CI/CD Pipelines
- **Protocols & Industrial Tech:** Modbus, PLC Communications, SCADA, IO Server Integration
- **Core Competencies:** RESTful API Design, JWT Authentication, Cloud Data Pipelines, Systems Architecture, Industrial Automation, Caching Strategies, Data Analytics and Visualization